

Lesson 8: Loops Make

45 minutes

Overview

Using Programming Patterns and a step-by-step approach students make their own version of a Lock Screen Maker app. At the beginning of the lesson students are able to explore a working version of the app. They are then given the design elements of the app but begin with minimal starting code. A progression of levels guides students on the high level steps they should use to develop their app but leaves it to them to decide how to write the code. At the end students submit their apps and answer a free-response question in the style of a Create performance task written response prompt. Student apps and written responses can be assessed using a provided rubric.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- ▶ AP - Algorithms & Programming

Agenda

Warm Up (2 minutes)

Intro the Project

Activity (38 minutes)

Build the Lock Screen Maker App

Wrap Up (5 minutes)

Assessment: Make Project

Objectives

Students will be able to:

- Implement programming patterns with loops to develop a functioning app
- Recognize the need for programming patterns with loops as part of developing a functioning app
- Use debugging skills as part of developing an app
- Write comments to clearly explain both the purpose and function of different segments of code within an app

Preparation

- Review the Free-Response Supports and consider which, if any, supports are suitable for your classroom.
- Check the "[Teacher's Lounge](#)" forum for verified teachers to find additional strategies or resources shared by fellow teachers
- If you are teaching virtually, consider checking our [Virtual Lesson Modifications](#)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- [CSP Unit 6 - Lesson 8 - Loops Make](#) - Slides [▼ Make a Copy](#)
- [Writing Support - Loops Make](#) - Handout [▼ Make a Copy](#)

- [**CSP Loops Make - Lock Screen**](#)

[**Maker**](#) - Activity Guide

▼ Make a Copy

Teaching Guide

Warm Up (2 minutes)

💡 Teaching Tip ▲

Short Intro: The Warm Up today is short and light. Students should spend the maximum amount of time working on their projects.

Intro the Project

🎤 Remarks

For the past few days, we've learned a lot about using lists to help apps keep track of information. In today's Make Project you'll be practicing using lists and programming patterns with loops to create a functioning Lock Screen app.

Activity (38 minutes)

Build the Lock Screen Maker App

Group: Make a determination as to whether this project will be completed in pairs or individually. You may even choose to let students decide.

Distribute: Make sure students have access to scrap paper and pencils / paper for drawing flow charts.

📖 **Level 1 - Explore:** Have students explore the working Lock Screen Maker app in Level 1. If students are not working in pairs they should still discuss with a neighbor.

📖 Discuss:

- *Where (if at all) do you think this app is using a loop?*
- *Where (if at all) do you think this app is using a list?*

Discussion Goals:

Where (if at all) do you think this app is using a list?

- It's more important that students justify their answers. However, students may recognize that the different icons are likely being chosen from a list.

Where (if at all) do you think this app is using a loop?

- Students will hopefully recognize that with 20 different icons to be updated by each button there is almost certainly a loop being used by each of those button.

Remarks

Now let's build the this app. The screen has been set up for you - it's your job to add the code!

Distribute: Give students copies of **CSP Loops Make - Lock Screen Maker** if you will be using it during the class.

Teaching Tip

Supporting Students: While students are working on their apps, circulate the room and check in with students who need a little help. Encourage students to collaborate and discuss bugs with each other.

Debugging: Review with students steps they can use to debug if they get stuck:

- Build small parts of the program at a time and test them to make sure they work
- Run the code on turtle mode
- Add the variables to the watcher
- Explain the code to a friend and clarify the differences between what they expect to happen and what is actually happening in

Do This: Direct students to level two where they complete the Lock Screen Maker app. Based on the needs of your classroom decide whether you will collectively go through the activity guide or have students complete it individually. Afterwards give them time to work on their projects and circulate the room to offer support. Students who finish early can work on the extensions suggested in the activity guide.

Submit: Encourage students to check the rubric on the last page of the Activity Guide before submitting.

Teaching Tip

Adapting Make Lessons - You may decide that you'd like to have your students write the code for a modified or entirely different app using the same skills. This might help you differentiate your instruction or provide you with a way to reassess students. Check out the [r customizemakelessons_how-to-guide/csp/2025] for guidance on how to get started with this.

Wrap Up (5 minutes)

Remarks

Great work on your Lock Screen Maker app today. Before wrapping up, we will practice writing for the style of questions that are asked as part of the Create performance task on the AP exam. Respond to the question in Step 5 in your activity guide, and submit it along with your project.

Do This: Complete the written response in Step 5 of the Activity Guide.

Teaching Tip

Each student, in every classroom, will come to the written response with a different level of comfort, ability, and familiarity with writing for these style of questions. As such, consider offering one or more of the following supports:

Writing Support: Provide students with these sentence starters for guiding their response.

App Interface Support: For students who did not fully complete their app, provide them with this snippet from the exemplar code.

Assessment: Make Project

Use the rubric provided with the project to assess student projects.

✔ Assessment Opportunity ▲

Student responses can be used as a formative assessment to evaluate their understanding of connections between user interface features and functionality. Review their explanations for clarity, accuracy, and alignment with the AP exam's expectations. The provided rubric can be used to guide this review, and to help provide students with feedback to guide their learning and improvement.



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